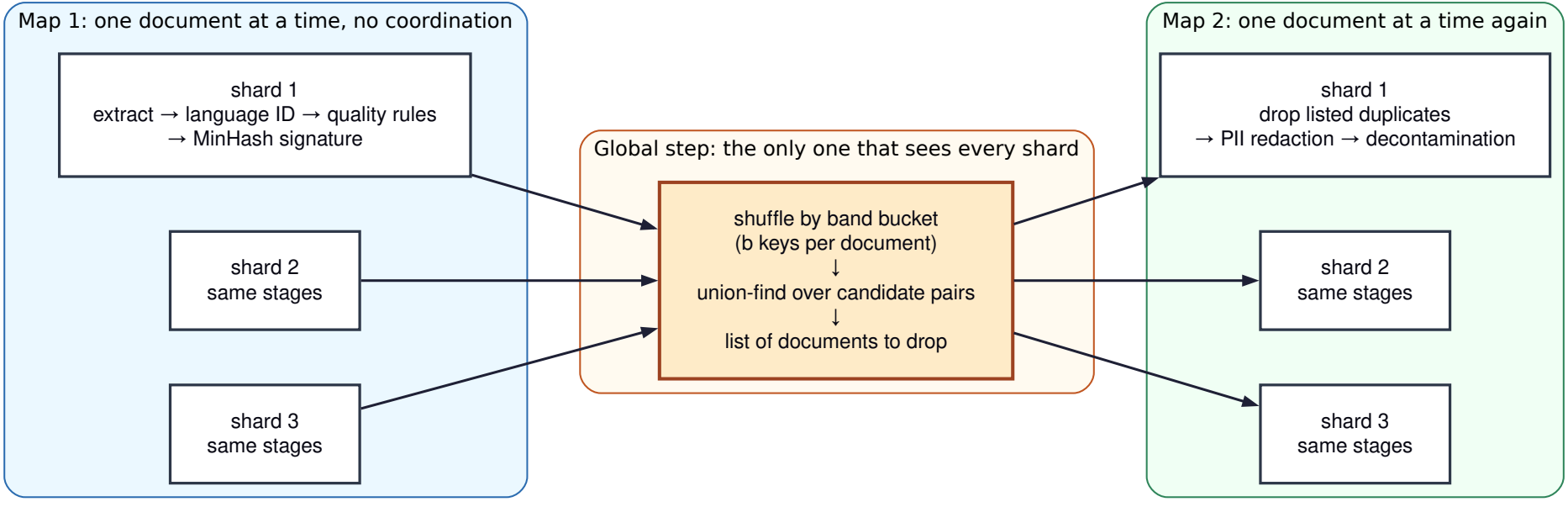
Map 1: one document at a time, no coordination
shard 1
extract → language ID → quality rules
→ MinHash signature
shard 2
same stages
shard 3
same stages
Global step: the only one that sees every shard
shuffle by band bucket
(b keys per document)
↓
union-find over candidate pairs
↓
list of documents to drop
Map 2: one document at a time again
shard 1
drop listed duplicates
→ PII redaction → decontamination
shard 2
same stages
shard 3
same stages